

In []:

PHYS 3316 — Quantum Mechanics I

Notebook 1 — The Infinite Square Well

Instructor: Eric R. Bittner

Department of Physics, University of Houston

Overview

The infinite square well is one of the most important exactly solvable models in quantum mechanics. Although the potential is highly idealized, it introduces many of the central ideas that recur throughout the course, including boundary conditions, energy quantization, stationary states, normalization, expectation values, uncertainty, and quantum superposition.

In this notebook you will derive the stationary-state solutions analytically, verify your results numerically using Python, and explore time-dependent superposition states that cannot easily be studied by hand.

Learning Objectives

After completing this notebook, you should be able to:

- Solve the time-independent Schrödinger equation for the infinite square well.
 - Apply boundary conditions to determine the allowed energy eigenvalues.
 - Normalize the stationary-state wavefunctions.
 - Compute expectation values and uncertainties for bound states.
 - Construct time-dependent solutions from stationary states.
 - Verify analytical results using numerical integration.
 - Visualize quantum states and probability densities using Python.
 - Explore arbitrary superposition states and their time evolution.
-

Required Reading

Griffiths & Schroeter,

Chapter 2

- Sections 2.1–2.2

Read these sections before beginning the notebook.

Software Requirements

This notebook uses

- Python 3
- NumPy
- SciPy
- Matplotlib

All calculations should be completed within the Jupyter notebook.

Submission Instructions

Complete **all analytical and computational problems**.

Before submitting,

1. Restart the notebook kernel.
2. Run **all** cells from beginning to end.
3. Verify that every figure and calculation is reproduced correctly.
4. Export the notebook as a PDF.

Submit your work using the filename

HW1_LastName_StudentID.pdf

through Canvas.

Honor Statement

You are encouraged to discuss concepts with your classmates, consult the course textbook, and use other legitimate resources.

You may also use AI-based tools (such as ChatGPT or GitHub Copilot) to assist your learning. However, all submitted work must represent **your own understanding**. You should be able to explain every derivation, equation, figure, and line of code contained in your submission.

Submitting work that you do not understand is considered a violation of the University's Academic Honesty Policy.

Notebook Organization

This notebook is organized into three parts.

Part I — Analytical Quantum Mechanics

Derive the stationary-state solutions of the infinite square well and calculate the corresponding physical observables.

Part II — Computational Quantum Mechanics

Verify the analytical results using numerical methods and visualize the wavefunctions and probability densities.

Part III — Beyond the Textbook

Use computation to investigate superposition states and time-dependent wave packets, illustrating quantum phenomena that extend beyond closed-form analytical solutions.

Problem 1 — The Infinite Square Well

A particle of mass m is confined to a one-dimensional infinite square well,

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & \text{otherwise.} \end{cases}$$

Inside the well, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x).$$

Outside the well,

$$\psi(x) = 0.$$

(a) General solution

Show that inside the well the Schrödinger equation can be written as

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0,$$

where

$$k^2 = \frac{2mE}{\hbar^2}.$$

Solve this differential equation to obtain the general solution

$$\psi(x) = A \sin(kx) + B \cos(kx).$$

(b) Apply the boundary conditions

The wavefunction must satisfy

$$\psi(0) = 0, \quad \psi(L) = 0.$$

Apply these boundary conditions to show that

$$B = 0$$

and that

$$kL = n\pi, \quad n = 1, 2, 3, \dots$$

Explain why the solution $n = 0$ is not physically acceptable.

(c) Quantized energies

Use your result for k to show that the allowed energies are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

Sketch the first four energy levels.

(d) Normalize the wavefunctions

The normalized wavefunctions satisfy

$$\int_0^L |\psi_n(x)|^2 dx = 1.$$

Determine the normalization constant and show that

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right).$$

(e) Orthogonality

The energy eigenfunctions form an orthonormal set.

Verify explicitly that

$$\int_0^L \psi_n(x) \psi_m(x) dx = 0, \quad n \neq m.$$

Evaluate the integral for

$$n = 1, \quad m = 2.$$

(f) Physical interpretation

Briefly answer the following questions.

1. Why are only certain energies allowed?
2. Why must the wavefunction vanish at the walls?
3. How does the infinite square well differ from a particle moving freely?

In []:

Problem 2 — A Quantum Particle in a Movable Piston

A particle of mass m is confined to a one-dimensional infinite square well of width L . Unlike the usual rigid box, imagine that the right-hand wall is a frictionless piston that can move in response to an external pressure P_{ext} .

The stationary-state energies are

$$E_n(L) = \frac{n^2 \pi^2 \hbar^2}{2mL^2}.$$

In thermodynamics, pressure is defined by

$$P = -\frac{dE}{dV},$$

where V is the system volume. For a one-dimensional box, we define the "volume" to be its length,

$$V = L.$$

Therefore, the quantum pressure exerted by the particle on the piston is

$$P_q = -\frac{dE_n}{dL}.$$

(a) Ground-state pressure

Differentiate the energy with respect to the box length and show that the pressure exerted by a particle in the n^{th} eigenstate is

$$P_q(n) = \frac{n^2 \pi^2 \hbar^2}{mL^3}.$$

Evaluate the pressure for the ground state ($n = 1$).

(b) Mechanical equilibrium

Suppose the piston is held in equilibrium by a constant external pressure P_{ext} .

If the particle occupies the ground state, the equilibrium box length L_1 satisfies

$$P_q(1) = P_{\text{ext}}.$$

Solve for the equilibrium length L_1 in terms of m , $\hbar$, and P_{ext} .

(c) Excitation of the particle

Suppose the particle is excited from the ground state ($n = 1$) to the first excited state ($n = 2$).

Immediately after the excitation, the piston has not yet moved.

1. By what factor does the quantum pressure increase?
 2. Is the piston pushed inward or outward?
 3. Explain your reasoning physically.
-

(d) New equilibrium

Eventually the piston moves until a new mechanical equilibrium is established,

$$P_q(2) = P_{\text{ext}}.$$

Determine the new equilibrium length L_2 .

Compute the expansion ratio

$$\frac{L_2}{L_1}.$$

(e) Excitation energy

The excitation energy at the original box length is

$$\Delta E_{21}(L_1) = E_2(L_1) - E_1(L_1).$$

After the piston expands to its new equilibrium position, the excitation energy becomes

$$\Delta E_{21}(L_2) = E_2(L_2) - E_1(L_2).$$

Calculate both quantities and determine the ratio

$$\frac{\Delta E_{21}(L_2)}{\Delta E_{21}(L_1)}.$$

(f) Interpretation

Briefly discuss why the excitation energy decreases after the piston expands, even though the particle remains in the $n = 2$ state.

How does this example illustrate the relationship between quantum mechanics and mechanics?

Problem 3 — Expectation Values and the Uncertainty Principle

Consider a particle in the n^{th} stationary state of the infinite square well,

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad 0 < x < L.$$

Recall that the expectation value of an observable $\hat{A}$ is

$$\langle A \rangle = \int_0^L \psi_n^*(x) \hat{A} \psi_n(x) dx.$$

(a) Position expectation value

The position operator is

$$\hat{x} = x.$$

Compute

$$\langle x \rangle = \int_0^L x |\psi_n(x)|^2 dx.$$

Interpret your result physically.

(b) Position uncertainty

Compute

$$\langle x^2 \rangle = \int_0^L x^2 |\psi_n(x)|^2 dx.$$

Then calculate the position uncertainty,

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}.$$

Simplify your final answer as much as possible.

(c) Momentum expectation value

The momentum operator is

$$\hat{p} = -i\hbar \frac{d}{dx}.$$

Compute

$$\langle p \rangle = \int_0^L \psi_n^* \left(-i\hbar \frac{d}{dx} \right) \psi_n dx.$$

Explain why your answer is consistent with the symmetry of the problem.

(d) Momentum uncertainty

Compute

$$\langle p^2 \rangle = \int_0^L \psi_n^* \left(-\hbar^2 \frac{d^2}{dx^2} \right) \psi_n dx.$$

Then determine

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}.$$

(e) Heisenberg uncertainty principle

Compute the uncertainty product

$$\sigma_x \sigma_p.$$

Verify that

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

Evaluate the uncertainty product numerically for

- $n = 1$
- $n = 2$
- $n = 5$

How does the uncertainty product change as the quantum number increases?

(f) Interpretation

Briefly discuss the following questions.

1. Why is the average position equal to the center of the box for every stationary state?
2. Why is the average momentum zero even though the particle has nonzero kinetic energy?
3. Why is the uncertainty product larger than the minimum value $\hbar/2$?

Problem 4 — Time Dependence and Stationary States

Consider a particle in the n^{th} energy eigenstate of the infinite square well,

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad 0 < x < L,$$

with energy

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}.$$

The corresponding time-dependent wavefunction is

$$\Psi_n(x, t) = \psi_n(x) e^{-iE_n t / \hbar}.$$

(a) Verify the time-dependent solution

Substitute $\Psi_n(x, t)$ into the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi,$$

and verify that it is a solution.

Use the fact that

$$\hat{H}\psi_n = E_n\psi_n.$$

(b) Probability density

The probability density is

$$\rho(x, t) = |\Psi_n(x, t)|^2.$$

Show that

$$|\Psi_n(x, t)|^2 = |\psi_n(x)|^2.$$

Explain why the phase factor

$$e^{-iE_nt/\hbar}$$

does not affect the probability density.

(c) Stationary states

A state whose probability density does not change with time is called a **stationary state**.

Explain why an energy eigenstate is stationary even though its wavefunction depends explicitly on time.

Does the word “stationary” mean that the particle is at rest?

(d) Time dependence of expectation values

For an operator $\hat{A}$ with no explicit time dependence, the expectation value is

$$\langle A \rangle(t) = \int_0^L \Psi_n^*(x, t) \hat{A} \Psi_n(x, t) dx.$$

Show that the time-dependent phase factors cancel and therefore

$$\langle A \rangle(t) = \int_0^L \psi_n^*(x) \hat{A} \psi_n(x) dx.$$

Conclude that expectation values of time-independent observables are constant in a stationary state.

(e) General superposition

A general state in the infinite square well may be written as

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

Assuming the eigenfunctions are orthonormal, show that normalization requires

$$\sum_{n=1}^{\infty} |c_n|^2 = 1.$$

What is the physical meaning of $|c_n|^2$?

(f) Two-state superposition

Consider the normalized state

$$\Psi(x, 0) = \frac{1}{\sqrt{2}} [\psi_1(x) + \psi_2(x)].$$

Write the state at time t as

$$\Psi(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x) e^{-iE_1 t/\hbar} + \psi_2(x) e^{-iE_2 t/\hbar}].$$

Compute the probability density $|\Psi(x, t)|^2$ and show that it contains an interference term proportional to

$$\cos\left(\frac{E_2 - E_1}{\hbar} t\right).$$

(g) Oscillation frequency

Define the angular frequency

$$\omega_{21} = \frac{E_2 - E_1}{\hbar}.$$

Using the infinite-well energy levels, show that

$$\omega_{21} = \frac{3\pi^2 \hbar}{2mL^2}.$$

Determine the corresponding oscillation period,

$$T_{21} = \frac{2\pi}{\omega_{21}}.$$

(h) Physical interpretation

Briefly answer the following questions.

1. Why is a single energy eigenstate stationary, while a superposition of energy eigenstates generally is not?
2. What produces the time-dependent interference term?
3. What determines the characteristic frequency of the probability-density oscillations?

Problem 5 — Time Evolution in Atomic Units

In Problem 4, you showed that a superposition of energy eigenstates is generally not stationary. In this problem, you will study the time evolution of a two-state superposition using **atomic units**.

Atomic units are convenient for calculations involving electrons because several fundamental constants are set equal to one:

$$\hbar = 1, \quad m_e = 1, \quad e = 1, \quad 4\pi\epsilon_0 = 1.$$

As a result, the Schrödinger equation and many quantum-mechanical expressions become considerably simpler.

The principal atomic units used in this problem are

Quantity	Atomic unit	SI value
Length	Bohr radius, a_0	$5.29177 \times 10^{-11} \text{ m}$
Energy	Hartree, E_h	$4.35974 \times 10^{-18} \text{ J}$
Energy	Hartree, E_h	27.2114 eV
Time	$\hbar/E_h$	$2.41888 \times 10^{-17} \text{ s}$

For an electron in a one-dimensional infinite square well, the energy levels in atomic units are

$$E_n = \frac{n^2\pi^2}{2L^2},$$

where L is expressed in Bohr radii and the energy is obtained in Hartrees.

For this problem, use

$$L = 20 a_0.$$

The normalized eigenfunctions are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad 0 < x < L.$$

Consider the two-state superposition

$$\Psi(x, 0) = \frac{1}{\sqrt{2}} [\psi_1(x) + \psi_2(x)].$$

At time t , the state is

$$\Psi(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x)e^{-iE_1t} + \psi_2(x)e^{-iE_2t}],$$

where both energy and time are expressed in atomic units.

(a) Why atomic units simplify the calculation

Begin with the usual phase factor,

$$e^{-iE_n t / \hbar}.$$

Explain why this becomes

$$e^{-iE_n t}$$

in atomic units.

What units must be used for E_n and t for this expression to be valid?

(b) Define the atomic-unit conversion factors

In a Python code cell, define the following constants:

```
bohr_to_meter = 5.29177e-11
hartree_to_joule = 4.35974e-18
hartree_to_ev = 27.2114
atomic_time_to_second = 2.41888e-17
```

Use these conversion factors to express the box length

$$L = 20 a_0$$

in

- meters,
- nanometers.

Compare this length with the 1 nm box used previously.

(c) Calculate the energy levels

Using

$$E_n = \frac{n^2 \pi^2}{2L^2},$$

calculate

$$E_1 \quad \text{and} \quad E_2$$

in Hartrees.

Convert both energies to electron volts using

$$1 E_h = 27.2114 \text{ eV}.$$

Display your results with appropriate labels and units.

(d) Calculate the oscillation period

The energy difference between the two states is

$$\Delta E_{21} = E_2 - E_1.$$

In atomic units, the angular frequency is

$$\omega_{21} = E_2 - E_1.$$

Explain why the factor of $\hbar$ does not appear in this expression.

The oscillation period is

$$T = \frac{2\pi}{\omega_{21}}.$$

Calculate T in atomic units of time and then convert it to

- seconds,
- femtoseconds.

Recall that

$$1 \text{ fs} = 10^{-15} \text{ s}.$$

(e) Construct the spatial grid

Create a NumPy array containing at least 500 equally spaced points over the interval

$$0 \leq x \leq L.$$

Because the calculation is performed in atomic units, the values of x and L should be entered directly in units of a_0 .

Define a Python function for the normalized infinite-well eigenfunction,

```
def psi_n(x, n, L):
```

```
    ...
```

Verify numerically that

$$\int_0^L |\psi_1(x)|^2 dx \approx 1$$

and

$$\int_0^L |\psi_2(x)|^2 dx \approx 1.$$

(f) Construct the time-dependent state

Define a Python function that returns

$$\Psi(x, t) = \frac{1}{\sqrt{2}} [\psi_1(x)e^{-iE_1t} + \psi_2(x)e^{-iE_2t}].$$

For example,

```
def Psi(x, t, L):
```

```
    ...
```

Remember that `numpy.exp` must be used to evaluate the complex exponential.

(g) Plot the probability density versus time

Evaluate the probability density

$$\rho(x, t) = |\Psi(x, t)|^2$$

at the following times:

$$t = 0, \quad t = \frac{T}{4}, \quad t = \frac{T}{2}, \quad t = \frac{3T}{4}, \quad t = T.$$

Plot all five probability densities on the same figure.

Use

$$\frac{x}{L}$$

for the horizontal axis so that the position runs from 0 to 1.

Clearly label each curve by its value of t/T .

Include

- axis labels,
- a legend,
- a descriptive title.

(h) Plot the real and imaginary parts

For the same five times, plot

$$\text{Re } \Psi(x, t)$$

and

$$\text{Im } \Psi(x, t).$$

Use separate figures for the real and imaginary parts.

Explain why the wavefunction may change even when some measurable quantities remain unchanged.

(i) Verify normalization during the evolution

For each of the five times, numerically evaluate

$$N(t) = \int_0^L |\Psi(x, t)|^2 dx.$$

Display the results in a table.

Verify that

$$N(t) \approx 1$$

at every time.

Explain why unitary time evolution preserves normalization.

(j) Interpretation

Briefly answer the following questions.

1. Why does the probability density oscillate even though each individual energy eigenstate has a time-independent probability density?
2. Which term in $|\Psi(x, t)|^2$ is responsible for the time dependence?
3. Why does the probability density return to its initial form at $t = T$?
4. What practical advantage did atomic units provide in this calculation?
5. Which conversion factors were required only when translating the final results back into familiar physical units?

Optional challenge — Animate the probability density

Create an animation of

$$|\Psi(x, t)|^2$$

over one complete period,

$$0 \leq t \leq T.$$

Use atomic units internally and display the current time as both

$$\frac{t}{T}$$

and femtoseconds.

To help you get going, I've written a starter code. SAVE THIS and understand how I structured it. It may be useful for future problems. I won't always give you starter code in future problem sets.

```
In [ ]: # =====
# PHYS 3316 – Quantum Mechanics I
#
# Notebook 1 : Infinite Square Well
# Computational Exercise:
#   Time Evolution of an Arbitrary Superposition
#
# Author:
# Date:
#
# Description
# -----
# This program constructs and visualizes the time-dependent
```

```

# wavefunction of a particle confined to a one-dimensional
# infinite square well. The wavefunction is represented as a
# linear combination of stationary energy eigenstates,
#
#        $\Psi(x,t) = \sum c_n \psi_n(x) \exp(-iE_n t),$ 
#
# where the expansion coefficients  $c_n$  may be chosen by the user.
#
# The program computes
#
#   • stationary eigenfunctions  $\psi_n(x)$ 
#   • energy eigenvalues  $E_n$ 
#   • time-dependent wavefunction  $\Psi(x,t)$ 
#   • probability density  $|\Psi(x,t)|^2$ 
#   • expectation values  $\langle x \rangle(t)$  and  $\langle p \rangle(t)$ 
#   • phase-space trajectory  $(\langle x \rangle, \langle p \rangle)$ 
#
# All calculations are performed in atomic units.
#
# Atomic Units
# -----
#        $\hbar = 1$ 
#        $m_e = 1$ 
#        $e = 1$ 
#
# Length   : Bohr radius ( $a_0$ )
# Energy   : Hartree ( $E_h$ )
# Time     :  $\hbar/E_h$ 
#
# Students are encouraged to modify:
#
#   • box length
#   • expansion coefficients
#   • number of basis states
#   • plotting parameters
#
# and investigate how these changes affect the quantum dynamics.
#
# =====

import numpy as np
import matplotlib.pyplot as plt

# =====

# USER PARAMETERS

# =====

# Infinite well width (Bohr radii)

L = 20.0

# Expansion coefficients
#

```

```

#  $\Psi(x,0) = \sum c_n \psi_n(x)$ 
#
# Example:
#
#   c = [1,1]           equal superposition (example)
#   c = [1,0]           ground state
#   c = [0,1]           first excited state
#   c = [1,1j]          relative phase  $\pi/2$ 
#
coefficients = np.array([
    1.0,
    1.0
], dtype=complex)

# Number of spatial grid points
Nx = 1000
# Number of time points
Nt = 500

# Example with relative phase:
#
# coefficients = np.array([
#     1.0,
#     1.0j
# ], dtype=complex)

# Times at which the wavefunction will be plotted.
# These are entered below after a reference period is calculated.

# =====
# ATOMIC-UNIT CONVERSIONS
# =====

hartree_to_ev = 27.211386
atomic_time_to_fs = 2.418884e-2
bohr_to_nm = 5.291772e-2

# =====
# INFINITE-SQUARE-WELL FUNCTIONS
# =====

def psi_n(n, x, L):
    """
    Normalized eigenfunction of the infinite square well.

    Parameters
    -----
    n : int
        Quantum number, n = 1, 2, 3, ...
    x : ndarray
        Position in Bohr radii.
    L : float
        Box length in Bohr radii.
    """

```

```

Returns
-----
ndarray
    Values of psi_n(x).
"""
return np.sqrt(2.0 / L) * np.sin(n * np.pi * x / L)

def energy_n(n, L):
    """
    Infinite-square-well energy in Hartrees.
    """
    return n**2 * np.pi**2 / (2.0 * L**2)

def normalize_coefficients(coefficients):
    """
    Normalize the expansion coefficients so that

        sum_n |c_n|^2 = 1.
    """
    norm = np.sqrt(np.sum(np.abs(coefficients)**2))

    if norm == 0:
        raise ValueError("At least one expansion coefficient must be nonzero")

    return coefficients / norm

# Normalize the user-supplied coefficients
coefficients = normalize_coefficients(coefficients)

# Number of states included in the expansion
number_of_states = len(coefficients)

# Quantum numbers n = 1, 2, ..., number_of_states
quantum_numbers = np.arange(1, number_of_states + 1)

# Corresponding energies
energies = np.array([
    energy_n(n, L) for n in quantum_numbers
])

def Psi(x, t, coefficients, energies, L):
    """
    Construct the time-dependent wavefunction

        Psi(x,t) = sum_n c_n psi_n(x) exp(-i E_n t).

    Atomic units are used, so hbar = 1.
    """
    wavefunction = np.zeros_like(x, dtype=complex)

    for n, c_n, E_n in zip(

```

```

        quantum_numbers,
        coefficients,
        energies
    ):
        wavefunction += (
            c_n
            * psi_n(n, x, L)
            * np.exp(-1j * E_n * t)
        )

    return wavefunction

# =====
# EXPECTATION VALUES AS FUNCTIONS OF TIME
# =====

# Number of time points used for the calculation
number_of_time_points = Nt

# Follow the system over one reference period
time_grid = np.linspace(
    0.0,
    reference_period,
    number_of_time_points
)

def expectation_x(x, wavefunction):
    """
    Calculate the expectation value <x>.

    Parameters
    -----
    x : ndarray
        Spatial grid in Bohr radii.
    wavefunction : ndarray
        Psi(x,t) evaluated on the spatial grid.

    Returns
    -----
    float
        Expectation value <x> in Bohr radii.
    """
    probability_density = np.abs(wavefunction)**2

    return np.trapezoid(
        x * probability_density,
        x
    )

def expectation_p(x, wavefunction):
    """
    Calculate the expectation value <p> using

```

```

        p = -i d/dx

in atomic units.
"""
dpsi_dx = np.gradient(
    wavefunction,
    x,
    edge_order=2
)

integrand = (
    np.conjugate(wavefunction)
    * (-1j * dpsi_dx)
)

p_average = np.trapezoid(
    integrand,
    x
)

# The exact expectation value is real because momentum is
# a Hermitian operator. A small imaginary component may remain
# because of numerical differentiation and integration.
return np.real(p_average)

# Arrays that will hold the expectation values
x_average = np.zeros_like(time_grid)
p_average = np.zeros_like(time_grid)

for index, t in enumerate(time_grid):

    wavefunction = Psi(
        x,
        t,
        coefficients,
        energies,
        L
    )

    x_average[index] = expectation_x(
        x,
        wavefunction
    )

    p_average[index] = expectation_p(
        x,
        wavefunction
    )

# =====
# DISPLAY THE STATE
# =====

```

```

print(f"Box length: {L:.3f} a0")
print(f"Box length: {L * bohr_to_nm:.3f} nm")
print()

print("Normalized expansion coefficients")
print("-----")

for n, c_n in zip(quantum_numbers, coefficients):
    probability = np.abs(c_n)**2

    print(
        f"n = {n:2d}: "
        f"c_n = {c_n.real:+.4f}{c_n.imag:+.4f}j, "
        f"|c_n|^2 = {probability:.4f}"
    )

print()
print(f"Sum of probabilities = {np.sum(np.abs(coefficients)**2):.6f}")

print()
print("Energy levels")
print("-----")

for n, E_n in zip(quantum_numbers, energies):
    print(
        f"E_{n} = {E_n:.6f} Hartree "
        f"= {E_n * hartree_to_ev:.4f} eV"
    )

# =====
# CHOOSE A REFERENCE TIME SCALE
# =====

# Identify the populated states
tolerance = 1.0e-12
populated = np.where(np.abs(coefficients) > tolerance)[0]

if len(populated) >= 2:
    populated_energies = energies[populated]

    # All positive energy differences among populated states
    energy_gaps = []

    for i in range(len(populated_energies)):
        for j in range(i + 1, len(populated_energies)):
            gap = abs(
                populated_energies[j]
                - populated_energies[i]
            )

            if gap > tolerance:
                energy_gaps.append(gap)

    smallest_gap = min(energy_gaps)

```

```

reference_period = 2.0 * np.pi / smallest_gap

print()
print(
    "Reference period based on the smallest populated "
    "energy gap:"
)
print(
    f"T_ref = {reference_period:.6f} atomic units"
)
print(
    f"T_ref = "
    f"{reference_period * atomic_time_to_fs:.6f} fs"
)

else:
    reference_period = 1.0

    print()
    print(
        "Only one energy eigenstate is populated, so the "
        "probability density is stationary."
    )

# =====
# TIMES TO PLOT
# =====

times = np.array([
    0.0,
    reference_period / 4.0,
    reference_period / 2.0,
    3.0 * reference_period / 4.0,
    reference_period
])

time_labels = [
    r"$t=0$",
    r"$t=T_{\rm ref}/4$",
    r"$t=T_{\rm ref}/2$",
    r"$t=3T_{\rm ref}/4$",
    r"$t=T_{\rm ref}$"
]

# =====
# SPATIAL GRID
# =====

x = np.linspace(0.0, L, Nx)

# =====
# PLOT THE INITIAL PROBABILITY DENSITY
# =====

```



```

initial_wavefunction = Psi(
    x,
    0.0,
    coefficients,
    energies,
    L
)

plt.figure(figsize=(8, 5))

plt.plot(
    x / L,
    np.abs(initial_wavefunction)**2
)

plt.xlabel(r"$x/L$")
plt.ylabel(r"$|\Psi(x,0)|^2$")
plt.title("Initial Probability Density")
plt.tight_layout()
plt.show()

# =====
# PLOT THE REAL PART
# =====

plt.figure(figsize=(8, 5))

for t, label in zip(times, time_labels):
    wavefunction = Psi(
        x,
        t,
        coefficients,
        energies,
        L
    )

    plt.plot(
        x / L,
        np.real(wavefunction),
        label=label
    )

plt.axhline(0.0, linewidth=0.8)
plt.xlabel(r"$x/L$")
plt.ylabel(r"$\mathrm{Re}\Psi(x,t)$")
plt.title("Real Part of the Wavefunction")
plt.legend()
plt.tight_layout()
plt.show()

# =====
# PLOT THE IMAGINARY PART
# =====

```

```

plt.figure(figsize=(8, 5))

for t, label in zip(times, time_labels):
    wavefunction = Psi(
        x,
        t,
        coefficients,
        energies,
        L
    )

    plt.plot(
        x / L,
        np.imag(wavefunction),
        label=label
    )

plt.axhline(0.0, linewidth=0.8)
plt.xlabel(r"$x/L$")
plt.ylabel(r"$\mathrm{Im}\backslash,\backslash\Psi(x,t)$")
plt.title("Imaginary Part of the Wavefunction")
plt.legend()
plt.tight_layout()
plt.show()

# =====
# PLOT THE PROBABILITY DENSITY
# =====

plt.figure(figsize=(8, 5))

for t, label in zip(times, time_labels):
    wavefunction = Psi(
        x,
        t,
        coefficients,
        energies,
        L
    )

    probability_density = np.abs(wavefunction)**2

    plt.plot(
        x / L,
        probability_density,
        label=label
    )

plt.xlabel(r"$x/L$")
plt.ylabel(r"$|\Psi(x,t)|^2$")
plt.title("Time Evolution of the Probability Density")
plt.legend()
plt.tight_layout()
plt.show()

```

```

# =====
# CHECK NORMALIZATION
# =====

print()
print("Numerical normalization check")
print("-----")

for t, label in zip(times, time_labels):
    wavefunction = Psi(
        x,
        t,
        coefficients,
        energies,
        L
    )

    normalization = np.trapezoid(
        np.abs(wavefunction)**2,
        x
    )

    print(
        f"t = {t:10.4f} a.u.: "
        f"integral |Psi|^2 dx = {normalization:.8f}"
    )

```

Box length: 20.000 a0

Box length: 1.058 nm

Normalized expansion coefficients

```

n = 1: c_n = +0.7071+0.0000j, |c_n|^2 = 0.5000
n = 2: c_n = +0.7071+0.0000j, |c_n|^2 = 0.5000

```

Sum of probabilities = 1.000000

Energy levels

```

E_1 = 0.012337 Hartree = 0.3357 eV
E_2 = 0.049348 Hartree = 1.3428 eV

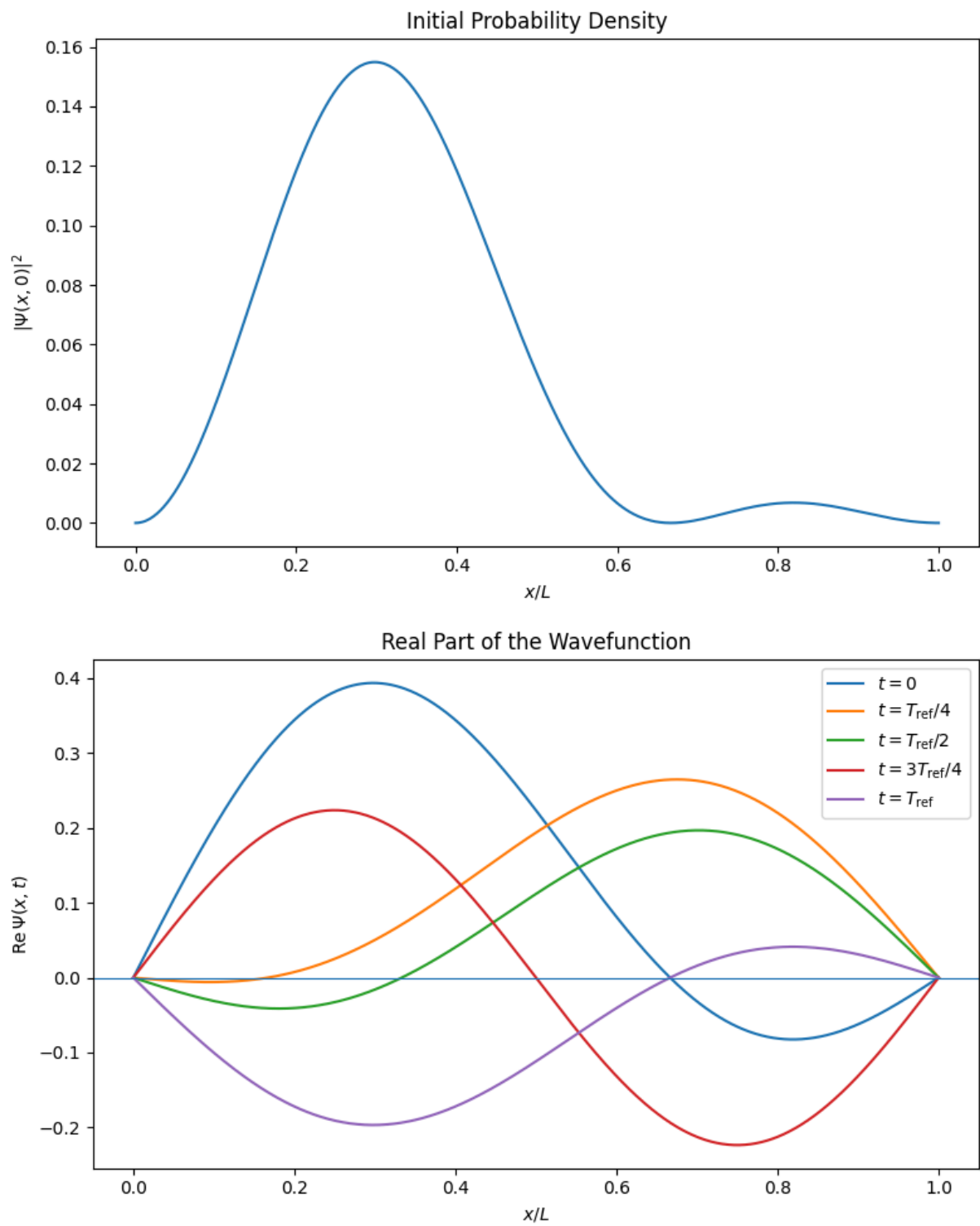
```

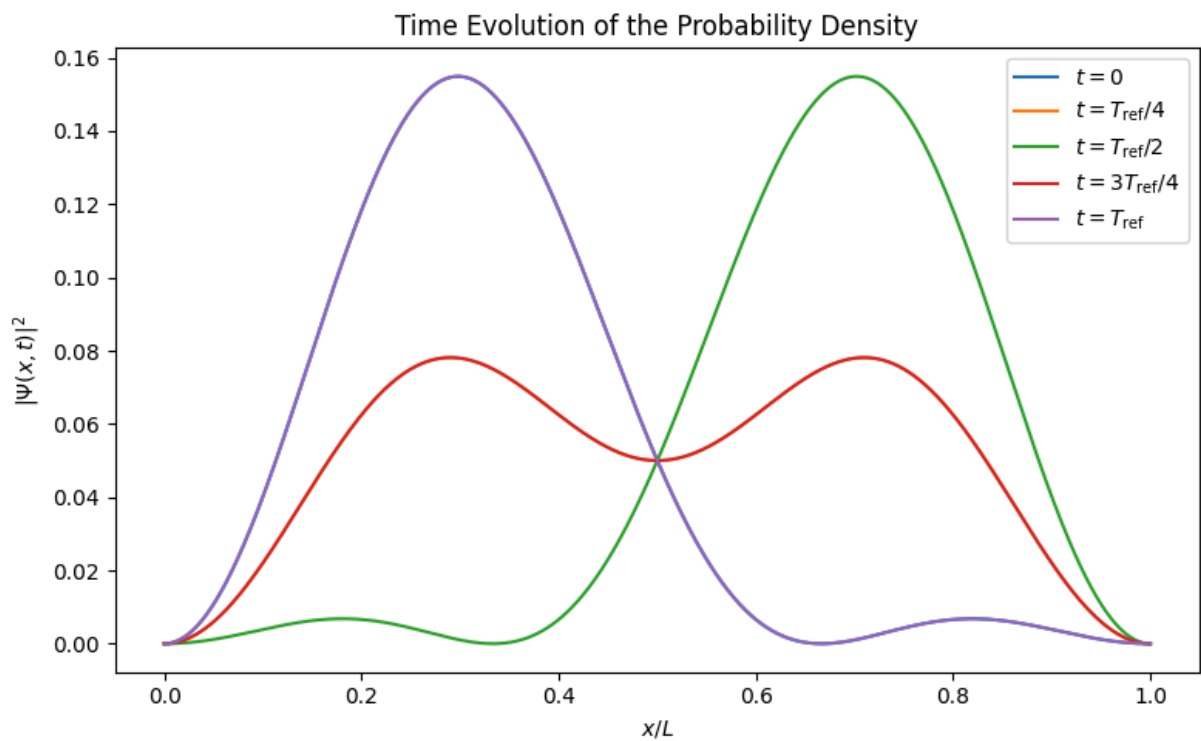
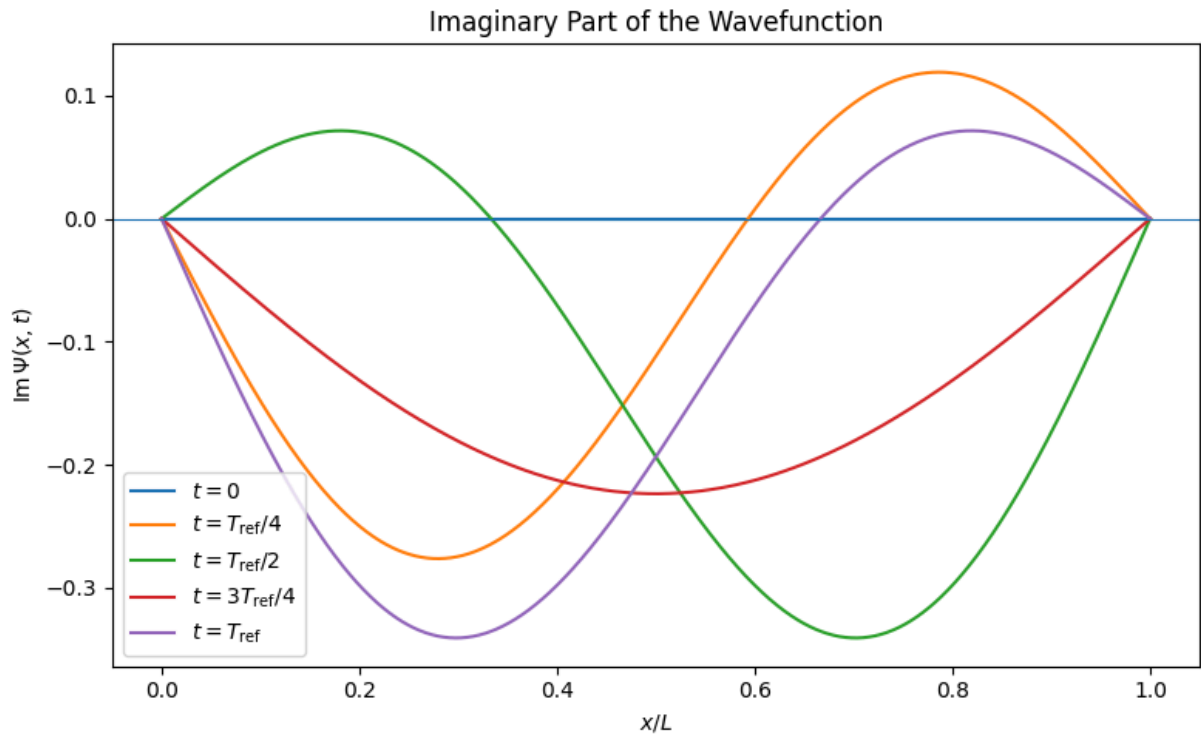
Reference period based on the smallest populated energy gap:

```

T_ref = 169.765273 atomic units
T_ref = 4.106425 fs

```





Numerical normalization check

```
-----
t =    0.0000 a.u.: integral |Psi|^2 dx = 1.00000000
t =   42.4413 a.u.: integral |Psi|^2 dx = 1.00000000
t =    84.8826 a.u.: integral |Psi|^2 dx = 1.00000000
t =   127.3240 a.u.: integral |Psi|^2 dx = 1.00000000
t =   169.7653 a.u.: integral |Psi|^2 dx = 1.00000000
```

Problem 6 — Expectation Values and Ehrenfest's Theorem

In the previous problem, you constructed a time-dependent wavefunction of the form

$$\Psi(x, t) = \sum_{n=1}^N c_n \psi_n(x) e^{-iE_n t}.$$

Although the probability density changes with time, the motion is not described by a definite particle trajectory. Instead, we can follow the time evolution of expectation values.

The expectation value of position is

$$\langle x \rangle(t) = \int_0^L \Psi^*(x, t) x \Psi(x, t) dx.$$

The expectation value of momentum is

$$\langle p \rangle(t) = \int_0^L \Psi^*(x, t) \left(-i \frac{\partial}{\partial x}\right) \Psi(x, t) dx,$$

where atomic units have been used so that $\hbar = 1$.

(a) Compute $\langle x \rangle(t)$

Using the wavefunction generated in the previous problem, numerically calculate

$$\langle x \rangle(t)$$

over at least one reference period.

Plot

$$\frac{\langle x \rangle(t)}{L}$$

as a function of

$$\frac{t}{T_{\text{ref}}}.$$

(b) Compute $\langle p \rangle(t)$

Use a numerical derivative to evaluate

$$\frac{\partial \Psi}{\partial x}.$$

Then calculate

$$\langle p \rangle(t) = \int_0^L \Psi^* \left(-i \frac{\partial \Psi}{\partial x} \right) dx.$$

Plot $\langle p \rangle(t)$ over the same time interval.

(c) Verify Ehrenfest's theorem

Use a numerical derivative to calculate

$$\frac{d\langle x \rangle}{dt}.$$

In atomic units, the electron mass is $m_e = 1$, so Ehrenfest's theorem predicts

$$\boxed{\frac{d\langle x \rangle}{dt} = \langle p \rangle.}$$

Plot both quantities on the same axes and compare them.

some helper routines have been added to the python code

(d) Interpretation

1. How is the oscillation of $\langle x \rangle$ related to the motion of the probability density?
2. At what points is $\langle p \rangle = 0$?
3. At what points is the magnitude of $\langle p \rangle$ largest?
4. Does the particle follow the curve $\langle x \rangle(t)$ as a classical trajectory?
5. How accurately does your numerical calculation satisfy Ehrenfest's theorem?

```
In [ ]: ###. This plots <x> vs t.

plt.figure(figsize=(8, 5))

plt.plot(
    time_grid / reference_period,
    x_average / L
)

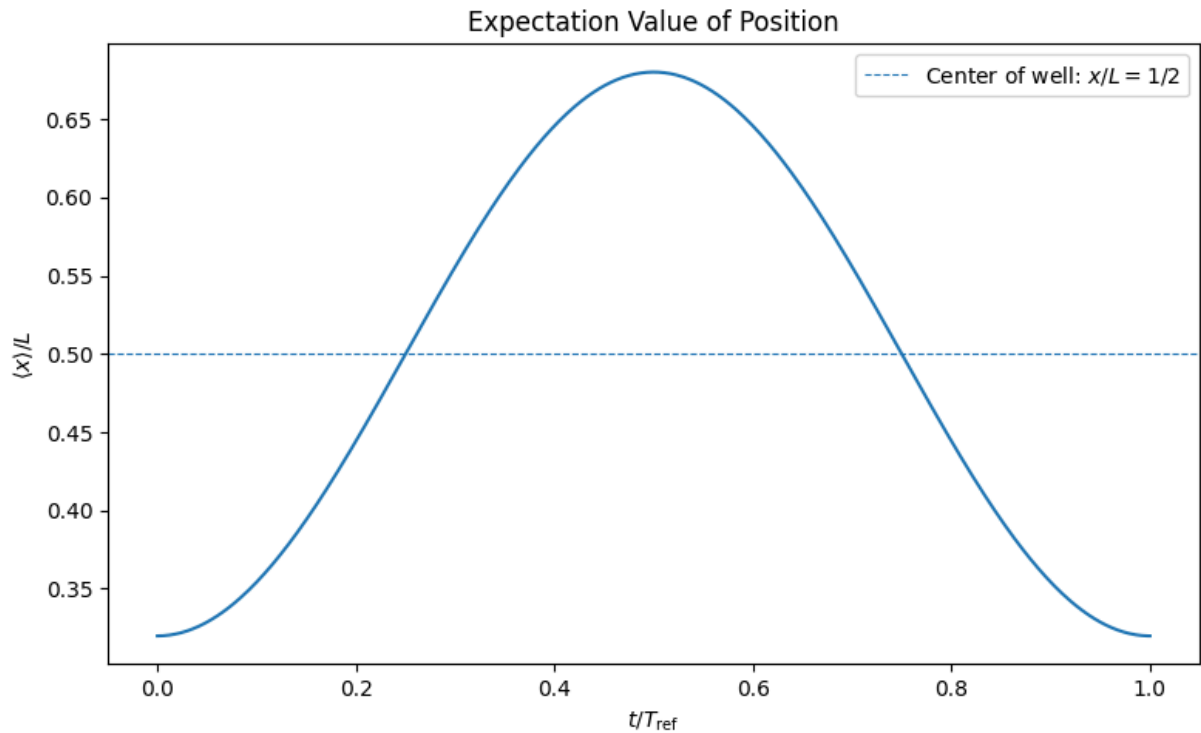
plt.axhline(
```

```

0.5,
linewidth=0.8,
linestyle="--",
label=r"Center of well:  $x/L=1/2$ "
)

plt.xlabel(r" $t/T_{\rm ref}$ ")
plt.ylabel(r" $\langle x \rangle / L$ ")
plt.title("Expectation Value of Position")
plt.legend()
plt.tight_layout()
plt.show()

```



In [4]: *## This plots $\langle p \rangle$ vs t .*

```

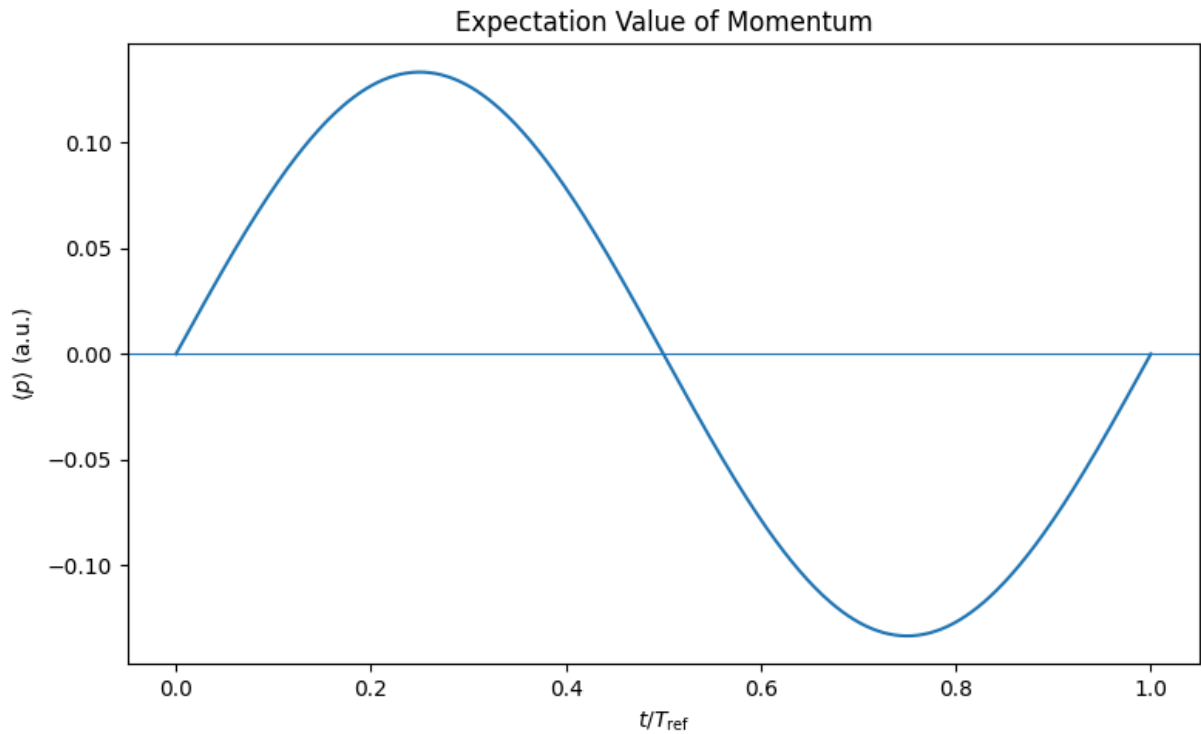
plt.figure(figsize=(8, 5))

plt.plot(
    time_grid / reference_period,
    p_average
)

plt.axhline(
    0.0,
    linewidth=0.8
)

plt.xlabel(r" $t/T_{\rm ref}$ ")
plt.ylabel(r" $\langle p \rangle$  (a.u.)")
plt.title("Expectation Value of Momentum")
plt.tight_layout()
plt.show()

```

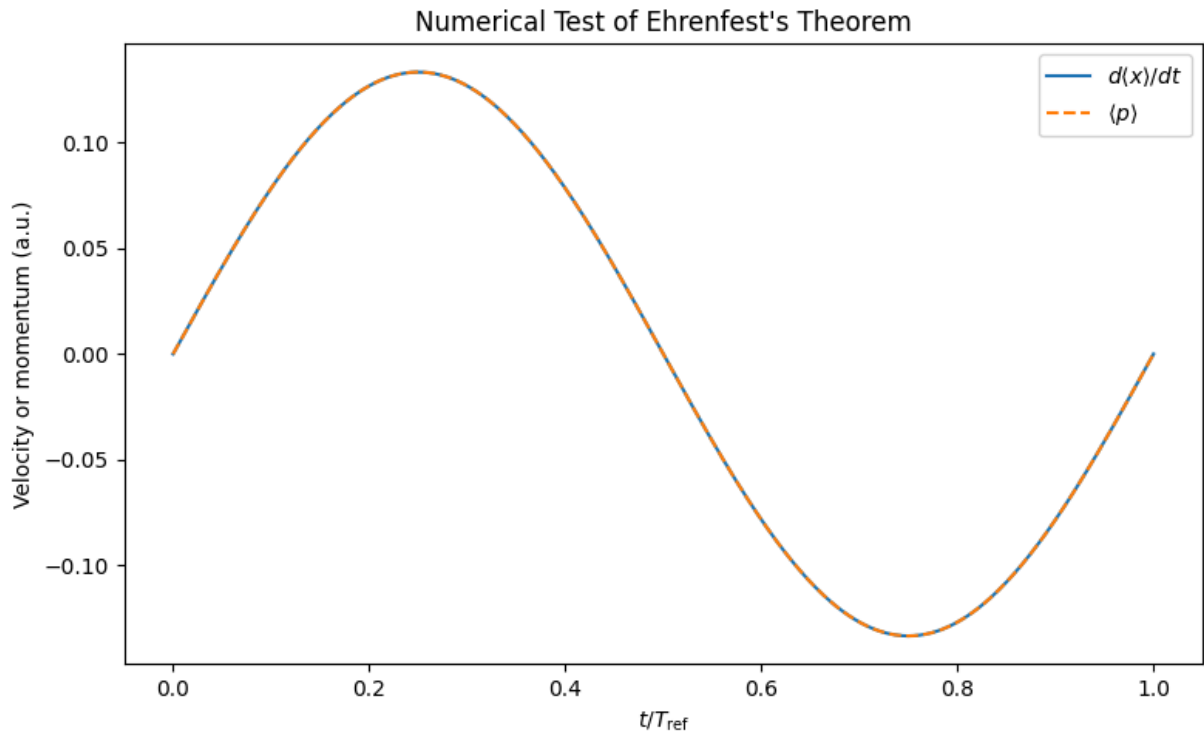
```
In [5]: # Numerical time derivative of <x>
dx_average_dt = np.gradient(
    x_average,
    time_grid,
    edge_order=2
)

plt.figure(figsize=(8, 5))

plt.plot(
    time_grid / reference_period,
    dx_average_dt,
    label=r"$d\langle x \rangle/dt$"
)

plt.plot(
    time_grid / reference_period,
    p_average,
    linestyle="--",
    label=r"$\langle p \rangle$"
)

plt.xlabel(r"$t/T_{\rm ref}$")
plt.ylabel("Velocity or momentum (a.u.)")
plt.title("Numerical Test of Ehrenfest's Theorem")
plt.legend()
plt.tight_layout()
plt.show()
```



7 - Classical vs. quantum phase space

In the plot below, I plot $\langle x \rangle(t)$ vs $\langle p \rangle(t)$ for the particle in a box for a state prepared in the superposition $\Psi(x, 0) = (\phi_1(x) + \phi_2(x))/\sqrt{2}$. What would you expect this to be for a purely *classical* particle with the same initial energy?

```
In [ ]: # =====
# PARAMETRIC PLOT: <p>(t) VERSUS <x>(t)
# =====

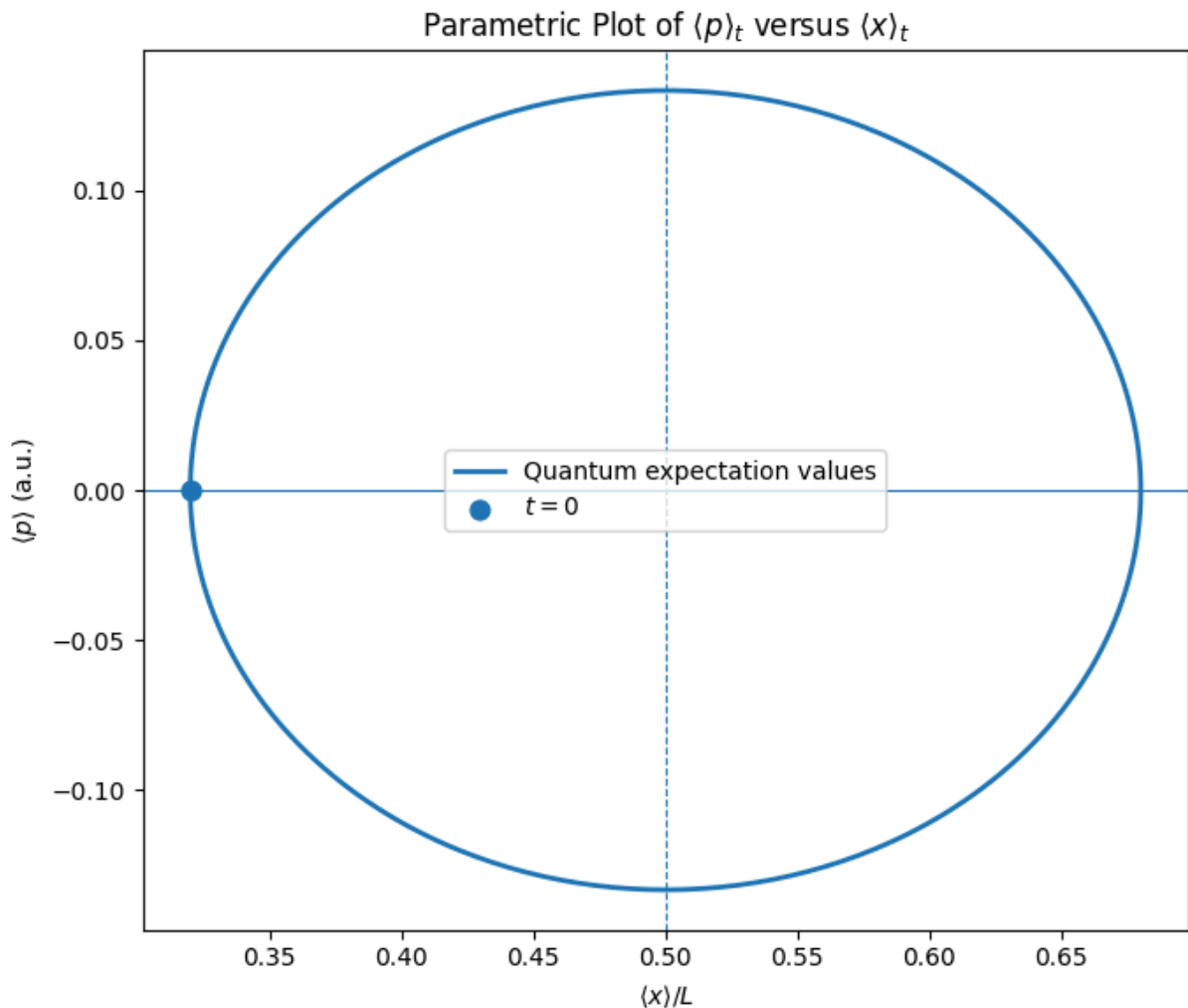
plt.figure(figsize=(7, 6))

plt.plot(
    x_average / L,
    p_average,
    linewidth=2,
    label="Quantum expectation values"
)

# Mark the initial point
plt.scatter(
    x_average[0] / L,
    p_average[0],
    s=60,
    label=r"$t=0$"
)

plt.axhline(0.0, linewidth=0.8)
plt.axvline(0.5, linewidth=0.8, linestyle="--")
```

```
plt.xlabel(r"$\langle x \rangle_t / L$")
plt.ylabel(r"$\langle p \rangle_t$ (a.u.)")
plt.title(r"Parametric Plot of $\langle p \rangle_t$ versus $\langle x \rangle_t$")
plt.legend()
plt.tight_layout()
plt.show()
```



In []:

Problem 8 (Challenge) — An Arbitrary Quantum State

Suppose the wavefunction for a particle confined to a one-dimensional infinite square well of length L is

$$\Psi(x, 0) = Ax(L - x), \quad 0 \leq x \leq L,$$

where A is a normalization constant.

Unlike the stationary states studied previously, this wavefunction is **not** an energy eigenstate. Instead, it represents a superposition of many stationary states.

(a) Normalize the wavefunction

Determine the normalization constant A .

Plot the normalized wavefunction on the interval

$$0 \leq x \leq L.$$

Which stationary state does this wavefunction most closely resemble? Explain your reasoning based on the shape of the wavefunction.

(b) Compute expectation values

Calculate

$$\langle x \rangle, \quad \langle x^2 \rangle, \quad \langle p \rangle, \quad \langle p^2 \rangle, \quad \langle H \rangle.$$

Compare these values with those of the exact ground state, $n = 1$.

Which quantities are nearly identical, and which differ the most?

(c) Expand the wavefunction in the stationary-state basis

Any wavefunction satisfying the boundary conditions of the infinite square well can be written as

$$\Psi(x, 0) = \sum_{n=1}^{\infty} c_n \psi_n(x),$$

where

$$c_n = \int_0^L \psi_n^*(x) \Psi(x, 0) dx.$$

Compute the coefficients c_n .

You may evaluate these integrals:

- analytically,
- symbolically using a package such as SymPy, or
- numerically.

Construct a table of the expansion coefficients.

State n	Coefficient c_n	Probability
1	---	---

State n	Coefficient c_n	Probability
2	---	---
3	---	---
...		

Continue adding states until

$$|c_n| \approx 10^{-5}.$$

Note, you'll need about 50 terms to converge the expansion coefficients to this accuracy! Do you notice something interesting in which coefficients are always $c_n = 0$

(d) Interpretation

1. Which stationary state has the largest probability?
2. Are both even and odd values of n present in the expansion? Explain your result using the symmetry of $\Psi(x, 0)$ about the center of the well.
3. Verify numerically that

$$\sum_{n=1}^{\infty} |c_n|^2 = 1.$$

4. Explain why a wavefunction that closely resembles the ground-state wavefunction is nevertheless not an exact ground state.

Optional Extension — Time Evolution

Use your expansion coefficients as input to the particle-in-a-box computational code developed earlier.

Construct the time-dependent wavefunction

$$\Psi(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t / \hbar}.$$

Investigate how the initially smooth wavefunction evolves in time.

Compute and plot:

- the probability density $|\Psi(x, t)|^2$,
- the expectation value $\langle x \rangle(t)$,
- the expectation value $\langle p \rangle(t)$, and

- the parametric trajectory $(\langle x \rangle, \langle p \rangle)$.

Compare this motion with the two-state superposition studied in the previous computational exercise.